

FINAL EXPLANATION: CHEMISTRY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CHEMISTRY GRADE 10 – SUB-STRAND 1.1 INTRODUCTION TO CHEMISTRY: FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

Think back to the chapati dough on the hot jiko in Lesson 1. In this Final Explanation you will use evidence from all ten lessons to answer our Driving Question: "How does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?" Write your answers in the space under each prompt. Use full sentences, name specific Kenyan examples from our lessons, and give reasons for your ideas. Where a prompt asks for a claim, evidence and reasoning, make sure you include all three parts. Take your time: there is no single word answer, we are marking how well you explain.

Part 1: What happened to the chapati dough?

In Lesson 1 you watched raw chapati dough change on a hot jiko into a firm, golden, aromatic chapati, and you could not turn it back into dough. (a) Define Chemistry and matter in your own words. (b) Explain why we say the change to the chapati is irreversible, and why Chemistry is a better science than a simple physical change idea for explaining it. (c) Compare it with the water particle changes you saw in the PhET States of Matter simulation. How is that change different?

(a) Chemistry is the study of matter, its properties, its composition and the changes it undergoes. Matter is anything that has mass and takes up space; the chapati dough, the flour, water, oil and salt in it, and the jiko's heat-affected air are all examples of matter or involve matter. (b) The change is irreversible because heat caused a chemical change: new substances formed, and bonds in the original matter broke and new ones formed. This is why the chapati has a different colour, aroma and firm texture, and why we can never get the soft dough back even though the ingredients were once all there. Chemistry is the science that studies the composition and changes of matter itself, so it explains this change. (c) In the PhET simulation, water particles moved from tightly packed (solid) to loosely spaced (gas) and back again. The particles stayed water, so the change was reversible at the particle level. In the chapati, new chemically different substances were made, so it cannot be reversed.

Part 2: Branches of Chemistry and the Maillard reaction

In Lesson 2 you studied seven Branch Cards. (a) Name the branch of Chemistry that best explains the browning of the

(a) Biochemistry best explains the browning, because it deals with the chemical reactions of substances such as proteins and sugars found in food and living things. (b) The Maillard reaction is a chemical reaction between amino acids (from wheat protein) and sugars (from wheat starch) driven by heat. It produces new molecules that give the chapati its brown colour, its aroma and its firm texture. Because new, chemically different molecules are formed, the change cannot be reversed. (c) Inorganic Chemistry:

chapati and say why. (b) Describe in simple terms what the Maillard reaction is and what it produced in the chapati. (c) Choose TWO other branches and give a Kenyan example for each of how that branch is used.	Bamburi Cement production uses inorganic materials. Analytical Chemistry: KEBS tests food to check that it is safe and of good quality. (Other correct examples from the Branch Cards, such as Industrial Chemistry or Nuclear Chemistry, are also acceptable if linked to a Kenyan context.)
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Part 3: Chemistry in agriculture, food, medicine and industry

Chemistry is at work long before and long after the chapati is cooked. (a) Using the ingredients of the chapati (wheat flour, water, oil, salt), explain TWO ways Chemistry is involved in Kenya's food chain, using examples from Lessons 3 and 4. (b) Choose one example from Lessons 4 or 5 (medicine, manufacturing, sport, energy, entertainment or the environment) and explain how it is similar to the chemical change in the chapati.	(a) First, in farming, fertilisers such as NPK or urea change soil chemistry so that maize and wheat grow well in places like the Rift Valley, and soil pH decides which crops grow well, for example tea in Kericho. Second, in food processing, preservatives such as sodium benzoate stop microbes causing spoilage, and fermentation, a chemical change, turns porridge into uji and milk into mursik. Food labels on Pembe flour or Bidco oil show preservatives, expiry dates and quality marks. (b) At EABL, fermentation is a controlled chemical change in which new substances are formed from the original matter and cannot easily be turned back. This is similar to the chapati, where heat forms new substances from dough. (Another acceptable answer: Eliud Kipchoge's muscles break down carbohydrates such as ugali or energy gels in a chemical change, forming new substances and releasing energy.)
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Part 4: Drugs, substance use and tobacco – a claim, evidence and reasoning

A friend says: "Shisha and tobacco are natural plants, so they are safe." Write a Claim–Evidence–Reasoning (CER) response. (a) State your claim. (b) Give evidence about tar, nicotine and carbon monoxide from Lesson 8. (c) Use your Chemistry knowledge to explain your reasoning, and link it to the chapati phenomenon. (d) Using Lesson 7, explain what a drug, a prescription and a dosage are, and why dosage matters.	(a) Claim: The statement is chemically incorrect. Being natural does not mean being safe, because tobacco contains chemicals that harm the body. (b) Evidence: Tar coats and destroys lung tissue. Nicotine acts on nerve receptors and creates chemical dependency. Carbon monoxide binds to haemoglobin and blocks oxygen transport in the blood. (c) Reasoning: Chemistry studies all matter, living and non-living. The chemical substances in tobacco interact with body matter at a molecular level, and much of the damage is largely irreversible, just as heat irreversibly changes chapati dough into something different that cannot return to dough. The origin of a substance, natural or man-made, does not decide whether it is safe; its chemical effects do. (d) A drug is a chemical substance that changes how the body works. A prescription is a written instruction from a qualified health worker for a particular medicine (for example, Metformin needs one while Panadol does not). Dosage is the amount to be taken and how often. Dosage matters because too much of a chemical substance causes harm, as the Panadol and Metformin labels warn. Substance abuse is using substances wrongly or in harmful ways.
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Part 5: Answering the Driving Question

Now answer the full Driving Question: "How does Chemistry explain what happens to the matter around us every day — and why does understanding Chemistry help us make safer, smarter choices in our lives?" Use at least THREE lessons as evidence. Include (a) how Chemistry explains the chapati, (b) how Chemistry connects to at least two areas of life (such as food, health, industry, careers or consumer protection), and (c) one example of a safer, smarter choice that a Kenyan could make because of Chemistry. Finally, say how your thinking has changed since your first prediction in Lesson 1.

Chemistry explains what happens to matter around us because it studies the composition, properties and changes of matter. The chapati changes irreversibly on the jiko because heat causes a chemical change (the Maillard reaction in Biochemistry): amino acids and sugars react to make new substances with a new colour, aroma and texture, so the dough cannot be recovered. The same idea applies elsewhere. In agriculture and food, fertilisers and soil pH affect crops in the Rift Valley and Kericho, and preservatives and fermentation change uji and mursik. In medicine and consumer safety, drugs interact with the body at a molecular level, and KEBS standards and product labels on Jogoo maize flour, Jik bleach and paracetamol give information on ingredients, expiry dates and dosage. Careers across these sectors are open to everyone, whatever their gender. Understanding Chemistry helps us make safer, smarter choices: for example, a person can read a paracetamol label and follow the correct dosage, or decide not to smoke because tar, nicotine and carbon monoxide cause damage that is largely irreversible. At the start I thought the chapati just got hard from being hot and did not see it as a chemical change; now I understand that new substances formed, and that Chemistry explains this in food, medicine and daily life. (Accept personal, honest comparisons with the learner's own Lesson 1 prediction.)

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of the anchoring phenomenon	Clearly explains the chapati change as an irreversible chemical change with new substances formed (including the Maillard reaction) and links it accurately to other examples throughout.	Identifies the chapati change as irreversible and chemical, with some explanation of why, and makes at least one link to another example.	Describes the chapati changing but does not explain why it is irreversible or does not link it to Chemistry.
Accurate Chemistry ideas and vocabulary	Uses terms such as matter, chemical change, branches of Chemistry, drug, prescription and dosage correctly and precisely.	Uses most terms correctly with minor gaps or imprecision.	Uses few key terms, or uses them incorrectly.
Evidence from the lessons and Kenyan contexts	Supports ideas with specific evidence from at least three lessons (e.g. Rift Valley fertilisers, KEBS, tobacco chemicals, Panadol and Metformin labels).	Gives some specific evidence from two or more lessons.	Gives general statements with little or no specific evidence.

Reasoning and CER structure	Presents a clear claim supported by evidence and logical reasoning that explains why the evidence supports the claim.	Gives a claim and evidence, with partial reasoning.	Gives a claim with little evidence or reasoning.
Safer, smarter choices and reflection	Gives a realistic example of a safe choice grounded in Chemistry and thoughtfully compares current understanding with the Lesson 1 prediction.	Gives a safe choice example and a basic reflection on learning.	Gives no clear example of a safe choice or no reflection.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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